Fine-Grained Analysis of Optimization and Generalization for Overparameterized Two-Layer Neural Networks

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The paper in one minute

Question: Why do neural nets fit *real* labels faster than random labels, and which data property predicts generalization?

Model/Method: A wide, randomly initialized two-layer ReLU network; its training predictions are approximated by a linear dynamical system controlled by a data-dependent kernel/Gram matrix.

Finding: Different eigen-directions of the kernel learn at different rates; label alignment explains training-speed variation. A data-dependent complexity term $y^\top (H^\infty)^{-1} y$ underlies a width-independent generalization bound.

Why we care: Zero training loss alone does not explain training speed or generalization. The label–kernel alignment adds information hidden by a single worst-case convergence rate.

1. From global convergence to *which directions learn fast?*

For a sufficiently wide two-layer ReLU network, the empirical output vector $u(t)$ evolves approximately according to a kernel-gradient update. If H^∞ is the limiting positive-definite Gram matrix, then the linearized model gives

$$\widetilde{u}_{t+1} = \widetilde{u}_t - \eta H^\infty (\widetilde{u}_t - y), \quad e_{t+1} = (I - \eta H^\infty) e_t,$$

where $e_t = \widetilde{u}_t - y$ and y is the training-label vector. A lower bound on the smallest eigenvalue ensures eventual contraction, but it hides how different label directions behave.

Worked example: fast and slow label components

Let the toy kernel and learning rate be

$$H^\infty = \begin{pmatrix} 2 & 0 \\ 0 & 0.2 \end{pmatrix}, \quad \eta = 0.25, \quad y = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \widetilde{u}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The residual update is $e_{t+1} = \mathrm{diag}(0.5, 0.95) e_t$, so

$$e_1 = \begin{pmatrix} -0.5 \\ -0.95 \end{pmatrix}, \quad e_2 = \begin{pmatrix} -0.25 \\ -0.9025 \end{pmatrix}, \quad \|e_t\|^2 = 0.5^{2t} + 0.95^{2t}.$$

The first label direction is learned quickly; the second persists. Thus the same optimizer and network can display very different loss curves depending on which eigen-directions the target labels occupy.

Interpretation: The paper explains why meaningful labels can train faster than randomized labels *in the observed settings*, through the spectrum of H^∞ and projections $v_i^\top y$; it does not claim every random labeling is slower for every dataset.

2. Spectral training law and generalization quantity

Let $H^\infty = \sum_i \lambda_i v_i v_i^\top$ with positive eigenvalues. With a zero initial output in the linearized proxy,

$$\|e_t\|^2 = \sum_{i=1}^{n} (1 - \eta\lambda_i)^{2t} (v_i^\top y)^2.$$

Theorem 4.1 shows that the actual neural-network prediction error stays close to this expression (with an additive tolerance), under sufficiently large width, small step size, kernel nondegeneracy, and high-probability initialization conditions. This sharper result improves on an analysis using only $\lambda_{\min}$.

Generalization: the inverse-kernel cost

Theorem 5.1 produces a bound depending on a data-dependent quantity of the form

$$\sqrt{\frac{2\, y^\top (H^\infty)^{-1} y}{n}} \quad \text{plus smaller confidence terms,}$$

under its sampling, label and loss assumptions. This term is *not* a guarantee that every fitted network generalizes. Its significance is that it does not explicitly grow with network width.

Worked intuition: when is a label expensive to learn?

For our toy $H^\infty = \mathrm{diag}(2, 0.2)$, the kernel cost for $y = (1, 1)$ is

$$y^\top (H^\infty)^{-1} y = 1/2 + 1/0.2 = 5.5.$$

For $y = (1, 0)$ it is only 0.5. Placing energy in a low-eigenvalue direction is expensive for the complexity bound and slow for optimization. Theorem 5.1 also contains assumptions and constants not reflected in this two-dimensional demonstration.

> **Separate these statements:** Theorem 4.1 concerns training *dynamics*; Theorem 5.1 concerns population/test performance under statistical assumptions. One does not follow automatically from the other.

Read Section 4 (Theorem 4.1), Section 5 (Theorem 5.1), Figures 1–2, and the data-dependent complexity discussion.

3. Main contributions, assumptions, limits

Contributions. A mode-by-mode, kernel-spectral description of training error; an explanation for slower fitting of randomized labels in studied datasets; a bound on generalization controlled by label/kernel alignment rather than raw parameter count; learning guarantees for a broad class of sufficiently smooth target functions in the paper's setting. **Assumptions.** Two-layer ReLU network with specified random initialization and fixed output signs, squared-error GD, sufficiently large width and small learning rate, a positive-definite limiting kernel H^{∞}, and additional distributional/label and bounded-loss conditions for generalization. **Limitations.** Kernel-regime approximation may not capture rich feature learning when weights move far from initialization. Empirical real-versus-random label comparisons are dataset-specific. This paper's generalization results are not general theorems for arbitrary deep networks and optimizers.

Cross-paper connections

Du and Allen-Zhu (2019) establish that sufficiently wide networks can fit training data; Arora et al. clarify *which directions fit first* and what controls a particular generalization bound. **Saxe (2014)** likewise decomposes learning into data modes, but in deep linear rather than wide ReLU kernel dynamics. **Zou–Gu (Ref 86)** sharpens worst-case convergence requirements, while Arora et al. add mode-specific predictions.

Quick reference

Concept	Interpretation
H^{∞}	Data-dependent limiting prediction-gradient Gram matrix.
λ_i, v_i	Kernel eigenvalues and orthogonal data modes.
$(1-\eta\lambda_i)^t$	Per-mode training-error decay in the kernel proxy.
$v_i^{\top} y$	How strongly the labels align with mode i.
$y^{\top}(H^{\infty})^{-1}y$	Data-dependent complexity controlling the bound.

Papers Master note

Arora et al. refine wide two-layer ReLU optimization analysis using the eigenstructure of a limiting Gram matrix: label components in high-eigenvalue modes converge faster, and $y^{\top}(H^{\infty})^{-1}y$ helps control a width-independent generalization bound under additional statistical assumptions.

Source: original ICML 2019 PDF, Sections 3–5, Theorems 4.1 and 5.1. Examples are pedagogical.